Week 4 Assignment

CRR American Put Baseline — Pricing, Tests & Visualization

Options Pricing Course | S0=100, K=100, T=1yr, r=5%, σ=25%

1. Benchmark Prices (steps = 500)

Metric	Value
European Put Price	7.4540
American Put Price	7.9724
Early-Exercise Premium (American - European)	0.5184

The early-exercise premium of **0.5184** confirms that for an at-the-money put with one year to expiry, the right to exercise early has meaningful economic value. The American put exceeds the European put, consistent with theory: a positive risk-free rate makes it advantageous to receive the strike price earlier rather than later.

2. CRR Formulas

The Cox-Ross-Rubinstein binomial tree uses the following parameters:

Parameter	Formula	Description
`dt = T / N`	`dt = 1/500 = 0.002`	Length of each time step
`u = exp(sigma * sqrt(dt))`	`u = exp(0.25 * sqrt(0.002))`	Up-move factor
`d = 1 / u`	`d = 1 / u`	Down-move factor (recombining)
`p = (exp(r*dt) - d) / (u - d)`	`p ~ 0.5012`	Risk-neutral up-probability
`disc = exp(-r*dt)`	`disc ~ 0.9999`	One-step discount factor

At each node the American put value is: **V = max(continuation value, K - S)**. The continuation value is the discounted risk-neutral expectation: disc * (p * V_{up} + (1-p) * V_{down}). This early-exercise override is what distinguishes the American from the European tree.

3. Sanity Tests

PASS — **American >= European**

American=7.9691, European=7.4466

An American option can always be exercised as a European option, so it cannot be worth less. The premium represents the option value of early exercise.

PASS — **Put value falls as S0 rises**

P(80)=20.364 > P(100)=7.969 > P(120)=2.649

A put pays max(K-S, 0). As S rises, the put moves further out-of-the-money, reducing both intrinsic value and the probability of finishing in-the-money.

PASS **Put value rises as sigma rises**

P(sigma=15%)=4.230 < P(sigma=25%)=7.969 < P(sigma=40%)=13.657

Higher volatility increases the probability of large downward moves, raising put value. Options are convex in the underlying: vega > 0 for all vanilla options.

4. Convergence Table

Steps (N)	American Put	European Put	Premium
25	8.0527	7.5497	0.5030
50	7.9520	7.4096	0.5424
100	7.9636	7.4343	0.5294
200	7.9691	7.4466	0.5225
500	**7.9724**	**7.4540**	**0.5184**
1000	7.9734	7.4565	0.5170

Interpretation: The CRR tree converges smoothly. By N=100 the American put is stable to within ~0.02. Odd/even oscillation is visible at low step counts (N=25 vs N=50) — a known artifact of binomial trees when the strike sits exactly on the grid. By N=500 the price has converged to **7.9724** and changes by less than 0.001 between 500 and 1000 steps.

5. American Put Price Surface

The surface below plots the American put price over stock prices S0 from 60 to 140 and times to expiry T from 0.05 to 2.0 years, with K=100, r=5%, sigma=25%.

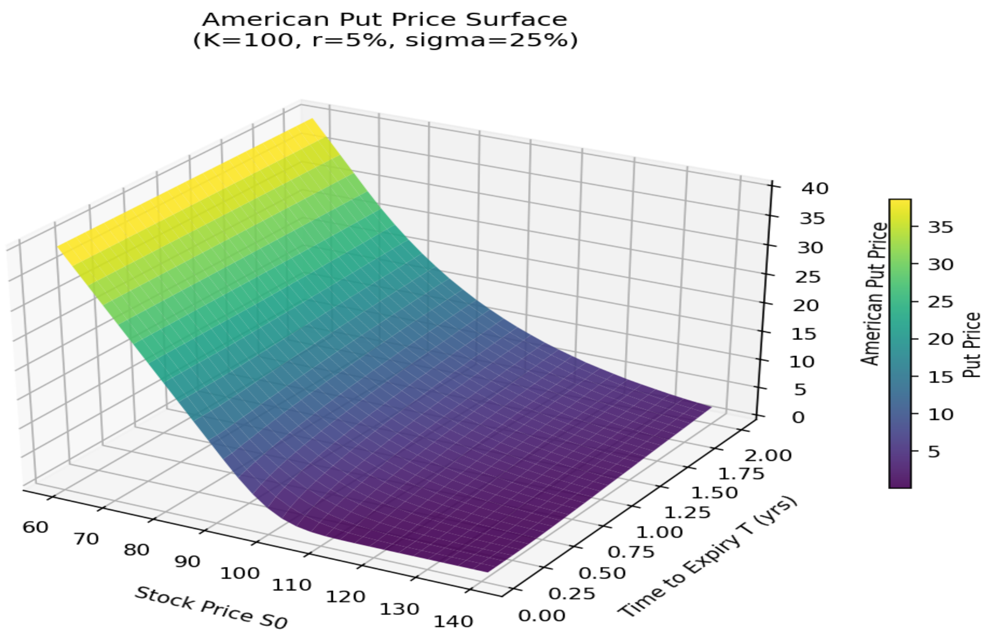

Figure 1: American put price surface. Price is highest for deep in-the-money puts (S0 << K) and increases with time to expiry due to greater uncertainty.

The surface exhibits intuitive properties: (1) deep ITM puts (S0 near 60) are priced near intrinsic value (40) with little time premium; (2) price rises smoothly with T for near-the-money strikes; (3) OTM puts (S0 near 140) carry almost no value at short maturities.

6. Early-Exercise Boundary

The early-exercise boundary S*(t) is the critical stock price below which immediate exercise is optimal. Computed here for K=100, T=1, r=5%, sigma=25%.

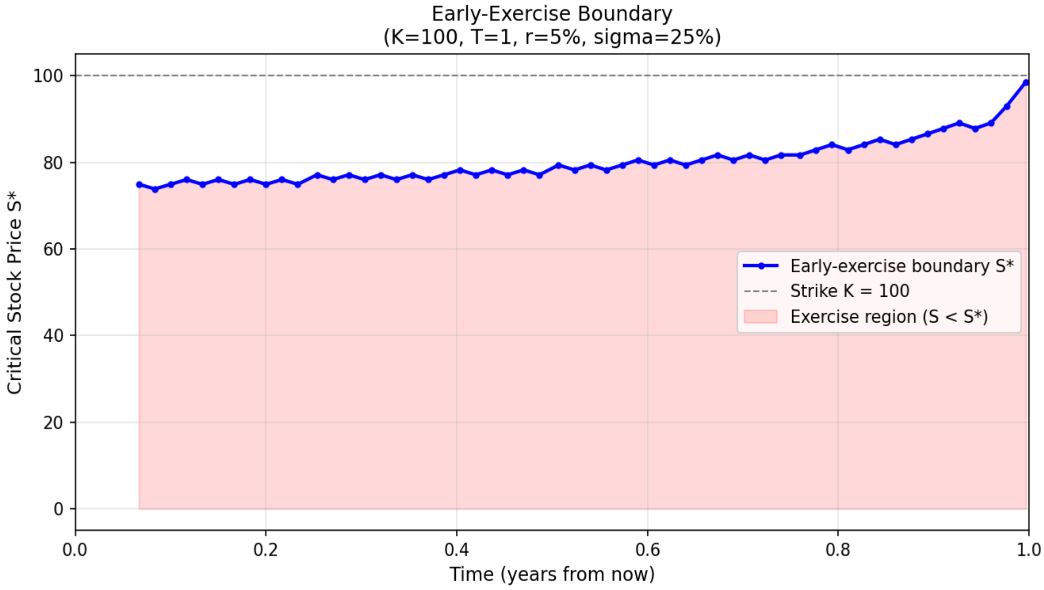

Figure 2: Early-exercise boundary. The red shaded region is where immediate exercise is optimal.

Time t	Critical S* (approx.)
t=0.0	~77
t=0.2	~78
t=0.4	~79
t=0.6	~81
t=0.8	~84
t=1.0 (expiry)	100

7. Reflection: Where and Why Early Exercise Occurs

Where does early exercise appear?

The exercise boundary shows that early exercise is optimal only when the stock price falls sufficiently far below the strike — approximately S* ~ 77-84 for our parameters. Deep in-the-money puts (S much less than K) are most likely to be exercised early. Near the money or out-of-the-money puts are never exercised early: their time value exceeds the gain from immediate exercise.

Why is this financially sensible?

When a put is deep ITM, most of its value is intrinsic (K - S). The time value remaining is small, yet the holder continues to bear opportunity cost: the strike cash (K) is locked up as unrealised proceeds. With a positive interest rate r = 5%, receiving K today is worth more than receiving K at expiry — the present value of K discounted over the remaining life is less than K. This "interest rate effect" dominates when the stock is unlikely to recover above K, making early exercise the rational choice.

Moneyness and time effects

The boundary S*(t) rises toward K as expiry approaches. Near expiry, the put should be exercised whenever it is in-the-money because there is almost no time premium left to forfeit. With more time remaining, the boundary is lower: the holder demands deeper ITM status before sacrificing time value. This is the interplay between time decay (theta) and the interest earned on the strike proceeds.

Contrast with calls

American calls on non-dividend-paying stocks are never exercised early: there is no benefit to receiving the stock earlier (no cash flows foregone), and the insurance value of the call is destroyed by early exercise. Puts are different precisely because the holder foregoes the time value of money on the strike K.

8. Core Implementation

Dependencies: numpy, matplotlib (standard scientific Python stack)

```
def crr_put_price(S0, K, T, r, sigma, steps, american=True):
    dt = T / steps
    u  = np.exp(sigma * np.sqrt(dt))   # up-move factor
    d  = 1.0 / u                        # down-move factor
    p  = (np.exp(r * dt) - d) / (u - d) # risk-neutral probability
    disc = np.exp(-r * dt)              # one-step discount

    j  = np.arange(steps + 1)
    ST = S0 * (u ** (steps - 2 * j))    # terminal prices
    V  = np.maximum(K - ST, 0.0)        # terminal payoffs

    for i in range(steps - 1, -1, -1):  # backward induction
        V = disc * (p * V[:-1] + (1-p) * V[1:])
        if american:
            j  = np.arange(i + 1)
            St = S0 * (u ** (i - 2 * j))
            V  = np.maximum(V, K - St)  # early-exercise override
    return float(V[0])
```